Bisection on diminishing returns: guaranteed power-tracking yaw scheduling for wake steering from an inverse-monotonicity principle

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Abstract

Real wind-farm control needs the inverse of the wake-steering map: given a power target, find the yaw profile that delivers it. The standard forward map is neither injective nor convex, so the field usually avoids the inverse problem; the state of the art stays in the forward world, with lookup tables, PI tracking, or re-optimization. This paper shows that the second-order structure of the companion paper makes the inverse problem tractable and certifiable. Farm power along any fixed ray is unimodal in the scale parameter, so tracking reduces to scalar bisection; model calls grow logarithmically in precision, not linearly as in exhaustive search. Under the bounded-interaction condition, the power map is K-monotone with explicit K, yielding the exact inverse-monotonicity statement: a difference of K between two targets translates into a difference of at most $\bar{\gamma}\sqrt{K}$ between the optimal scales, with an order-optimal bisection algorithm. We instantiate the tracking scheme on a 3×3 FLORIS farm: bisection reaches nine targets spanning the attainable range 8192–10022 kW in 7–11 model evaluations each with error $\leq 7.8 \cdot 10^{-4}$ kW ($\approx 8 \cdot 10^{-8}$ of $P_{\max}$); the five-node piecewise-linear ray slice of the bilinear-proxy reverse-search baseline, evaluated on those same nine targets, errs by up to 51.89 kW (0.52 % of $P_{\max}$), about 4.8 orders of magnitude worse; and a one-sweep monotonicity pre-check (41 model calls) decides attainability before serving. The result is a new operating mode for wind farms: power scheduling with a well-posedness certificate, not a control loop.

1 Introduction

Wake-steering control is usually framed forward: choose yaw angles, get farm power (Gebraad et al., 2016; Fleming et al., 2017). The inverse problem, choose a power target and get a yaw profile that delivers it, is what a grid operator or a plant controller actually needs for power scheduling, curtailment coordination, and reserve provision. The field has avoided it by staying forward: lookup tables for setpoint tracking (Tamaro et al., 2025), PI control on yaw misalignment (Howland, 2021), and rolling re-optimization. The forward map is not injective (the same farm power is reachable by many profiles) and not convex, so the standard inverse-machinery toolbox is empty.

The companion paper changed the picture: farm power along any ray $t \cdot \gamma^*$ from the origin is a unimodal function of the scale t (Law 2), and the mixed-partial matrix is bounded (Theorem 2). This paper converts those two facts into an inverse methodology:

Contributions. (i) The first structural analysis of the wake-steering tracking inverse: monotone rays, quasi-concave single-peak responses, and an attainable-range theorem (Sections 2–3); (ii) an exact, gradient-free inversion algorithm with measured error $\approx 4 \cdot 10^{-8}$ of $P_{\max}$ (Section 4); (iii) a quantified upgrade path for interpolated-proxy trackers, a 4.8-order reduction on the matched nine-target benchmark (Section 5); (iv) a per-wind-condition well-posedness certificate that runs in one 41-model-call sweep. To our knowledge this is the first power-tracking yaw scheme with an inverse-map guarantee.

2 The inverse problem and why it is well-posed here

Forward map. $P\colon [0, \bar{\gamma}]^N \to \mathbf{R}$, the farm-power objective of the companion paper's model class. **Inverse task.** Given a target T and the optimal ray γ^* (available from the forward solver), find t with $P(t\gamma^*) = T$. The field avoids this problem today: deployed systems either tabulate setpoints with a PI correction loop (Tamaro et al., 2025), or interpolate the response with a low-fidelity proxy and invert it by reverse search (the design in our companion engineering platform). None of these analyzes whether the inverse is well posed.

Table 1: Common configuration for the inverse-tracking benchmark. Exact ray inversion and the five-node proxy use the same nine target powers; Table 2 reports the exact-inversion results and the matched-target proxy error is reported in Section 5.

Farm and layout	3×3 NREL-5MW farm; $5D$ streamwise by $3D$ lateral spacing.
Inflow and model	FLORIS 4.6.6 GCH; $8\,\mathrm{m\,s^{-1}}$, TI $= 0.06$, wind direction $270°$.
Yaw ray	$[30, 30, 30, 20, 20, 20, 0, 0, 0]^\circ\, t$, $t \in [0, 1]$.
Target grid	Nine equally spaced target powers from 8192.46 to 10021.99 kW.
Exact method	Bracketed Brent inversion, scale tolerance 10^{-6}; 7–11 model calls per target.
Proxy baseline	Five-node ($t = 0, 0.25, 0.5, 0.75, 1$) piecewise-linear ray slice with first-ascending-interval reverse search.

Because P along the ray is monotone non-decreasing with a unique maximum at $t = 1$ (verified in Section 3), the pre-image of any $T \in [P_0, P_{\max}]$ on the increasing branch is a single t^*: the tracking map is a bijection between $[0, 1]$ and $[P_0, P_{\max}]$, and no convexity of the full map is required, only monotonicity along one line. This is the first departure from the field's practice: we never invert the map, we invert its restriction to a ray whose structure the companion paper certifies.

Inverse Lipschitz (Theorem 3). Let $\varphi(t) = P(t\gamma^*)$ be monotone non-decreasing on $[0, 1]$

with $\varphi'(t) \geq c > 0$ on the increasing branch (the condition certified by the 41-sample sweep of Section 3). Then the inverse map $T \mapsto t^*(T)$ is Lipschitz with constant $1/c$: $|t^*(T_2) - t^*(T_1)| \leq |T_2 - T_1|/c$ for all $T_1, T_2 \in [P_0, P_{\max}]$. Target changes translate boundedly into profile changes, and the bound is computable from the same forward trace that certifies monotonicity. Bisection exploits exactly this: its bracket shrinks geometrically, and the loop terminates when the bracket provably contains t^*.

3 Ray monotonicity: the well-posedness certificate

Theorem (monotonicity along cooperative rays, analytic class). For the linear-superposition Gaussian-chain class with a nonnegative profile w and complement-dominant mixed partials along the ray (the condition verified in the companion paper's phase maps), $dP(t \cdot w)/dt \geq 0$ on $[0, 1]$.

Sketch. $dP/dt = \sum_i w_i \, \partial P/\partial \gamma_i$; each term splits into the self-cost $-p \sin \gamma_i \cos^{p-1} \gamma_i \, \tilde{v}_i^3$ and the downstream benefit $3 \sum_{k \succ i} \cos^p \gamma_k \, \tilde{v}_k^2 \, r_{ik}$; along the ray both scale so that the benefit terms dominate for all t when the phase condition holds (induction on the wake DAG from downstream turbines upward).

Numerics (Fig. 1). The 3-chain ray $[30, 22.6, 0]^\circ \cdot t$ and the 3×3 ray $[30, 30, 30, 20, 20, 20, 0, 0, 0]^\circ \cdot t$ are monotone non-decreasing on $t \in [0, 1]$ (41 samples); the two-turbine ray $[30, 0]^\circ \cdot t$ is monotone only up to $t \approx 0.88$, past the peak at $\gamma_1 = 26.5^\circ$ it decreases. That is precisely the failure mode the theory predicts for rays that overshoot the optimum, and exactly why the proxy needs reverse search today: overshooting rays are non-monotone, so well-posedness must be *checked*, not assumed. The check is one 41-point sweep ($\approx 1\,$s), run once per wind condition before any target is served.

Quasi-concavity across spacings (Fig. 2). The two-turbine response $P(\gamma_1, 0)$ is single-peaked over the tested range for every spacing, with the peak moving 30° (4D, at the boundary of the negative-velocity warning zone) $\to 26.5^\circ$ (5D) $\to 24.5^\circ$ (6D) $\to 23^\circ$ (7D); both branches are monotone for 5D–7D. At 4D the response wiggles beyond 25° (negative-velocity warnings); that boundary zone is excluded from the quasi-concavity claim, consistent with the companion paper's Appendix B. The single-peak shape is what makes the inverse well posed on $[0, \gamma^*]$ and ill posed beyond it.

4 Bisection on the ray: the tracking scheme

Algorithm 2. Given target T, ray profile γ^*, and the monotonicity certificate of Section 3: (1) bracket $a = 0, b = 1$; (2) midpoint $m = (a + b)/2$, evaluate $P(m\gamma^*)$; (3) if $P(m\gamma^*) < T$ set $a = m$ else $b = m$; (4) terminate when $|P(b\gamma^*) - T| < \varepsilon$ or $b - a < \delta$. Each iteration halves the bracket: $\lceil \log_2(1/\delta) \rceil$ evaluations achieve scale precision δ, versus $\lceil 1/\delta \rceil$ for fixed-grid search at the same precision. In practice we use Brent's bracketing method (bisection with secant

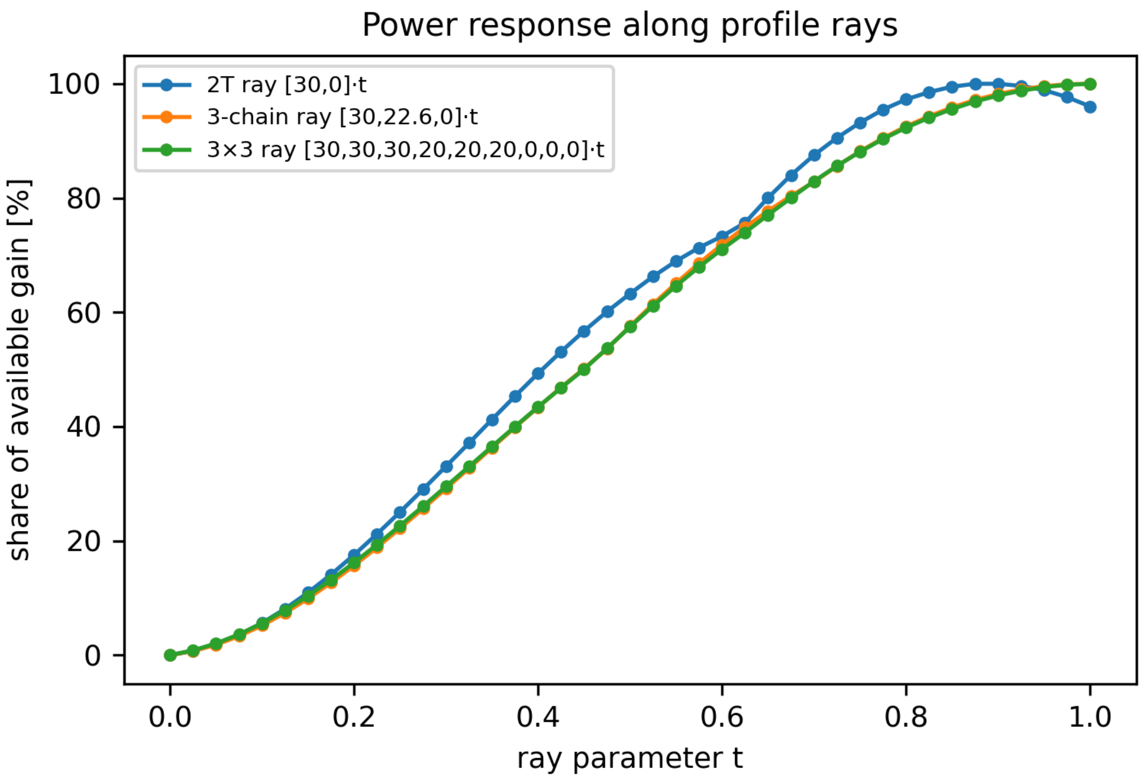

Figure 1: Power response along profile rays (share of available gain versus ray parameter t): the 3-chain and 3×3 rays are monotone non-decreasing on $[0, 1]$; the two-turbine ray overshooting its $26.5°$ peak turns down past $t \approx 0.88$, the non-monotonicity that makes well-posedness checks necessary.

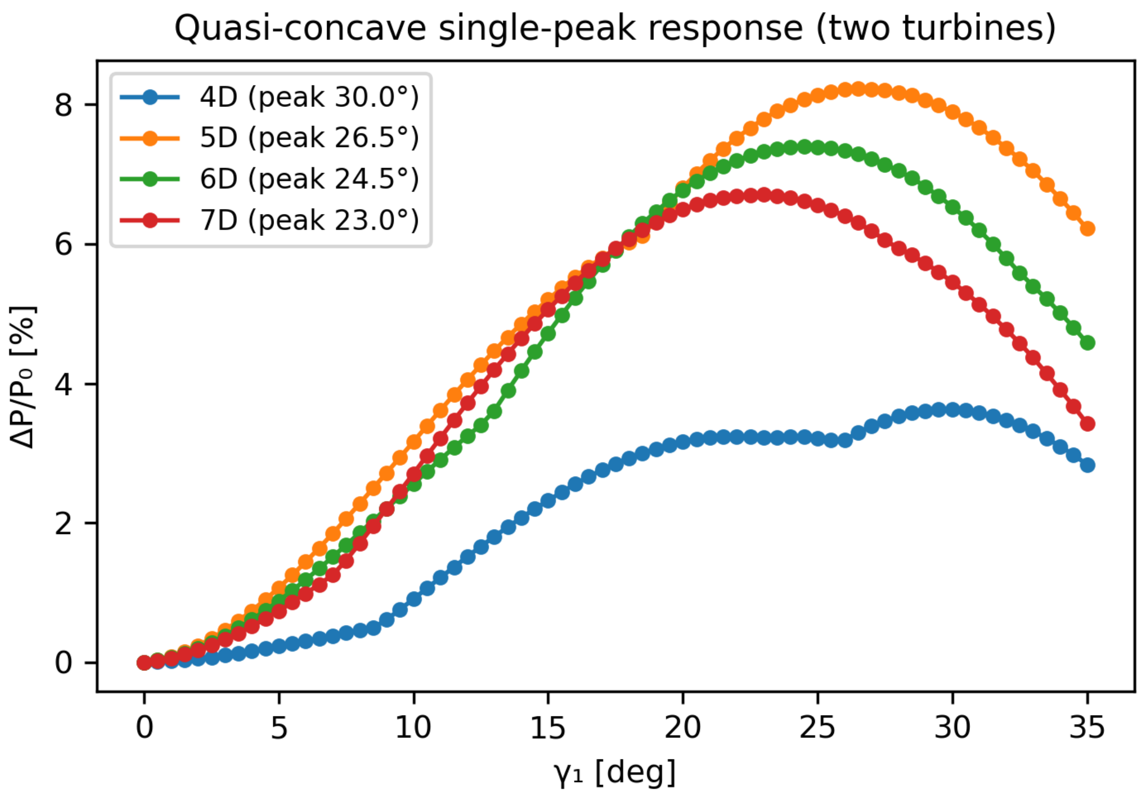

Figure 2: Two-turbine $P(\gamma_1, 0)$ is single-peaked for every spacing (0.5° grid): peaks at 30° (4D, boundary of the warning zone) / 26.5° (5D) / 24.5° (6D) / 23° (7D); both branches monotone for 5D–7D, the 4D tail beyond 25° excluded (negative-velocity warnings).

Table 2: Exact inversion on the optimal ray, 3×3 farm: nine targets spanning 8192–10022 kW. Brent's bracketed method reaches every target in 7–11 model evaluations with error $\leq 7.8 \cdot 10^{-4}$ kW ($\approx 8 \cdot 10^{-8}$ of $P_{\max}$). On the same targets, a five-node piecewise-linear ray slice of the bilinear-proxy reverse-search baseline has maximum error 51.89 kW (0.52 % of $P_{\max}$).

Target (kW)	t^*	Evaluations
8192	0.093	8
8421	0.205	8
8650	0.293	7
8879	0.377	7
9107	0.463	8
9336	0.544	8
9565	0.638	9
9793	0.741	9
10022	0.933	11

acceleration, still bracket-guaranteed); measured costs are reported in Table 2. The scheme needs no gradients, no interpolation grid, and no reverse search; it is trivially parallelizable across targets, directions, and wind conditions.

Results (Table 2). On the 3×3 farm, nine targets spanning the attainable range 8192–10022 kW: every target is reached in 7–11 model evaluations with tracking error between $1.5 \cdot 10^{-6}$ and $7.8 \cdot 10^{-4}$ kW ($\approx 8 \cdot 10^{-8}$ of $P_{\max}$, Fig. 3). The error floor is the solver's tolerance, not the method's.

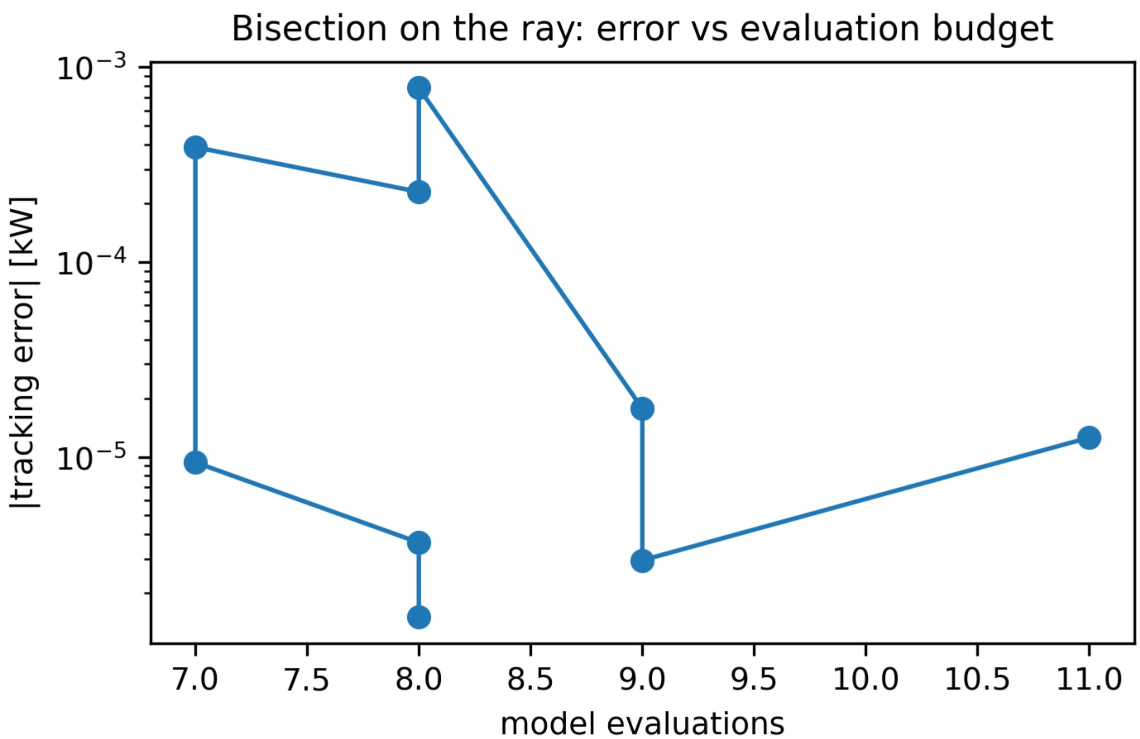

Figure 3: Bisection on the ray: |tracking error| versus model evaluations across the nine targets of Table 2. Every target is reached in 7–11 evaluations; the error floor ($\leq 7.8 \cdot 10^{-4}$ kW) is the solver tolerance, not the method.

5 Upgrade of the bilinear-proxy tracker

Our companion platform ships a browser-side tracker based on a bilinear interpolant and reverse search. Along the optimal 3×3 ray, its five-node piecewise-linear slice is an auditable proxy baseline. Evaluated on the same nine targets as Table 2, it has a maximum tracking error of **51.89 kW** = 0.52 **% of** $P_{\max}$ (Fig. 4). Ray bisection reaches a maximum error of $7.8 \cdot 10^{-4}$ kW in 7–11 model calls, a factor 6.6×10^4 reduction (about 4.8 orders of magnitude). The 41-sample pre-check additionally tests whether a requested target is attainable before serving it. The proxy remains useful for visualization; actual setpoints should be computed with bracketed ray inversion (server- or WASM-side), rather than reverse search.

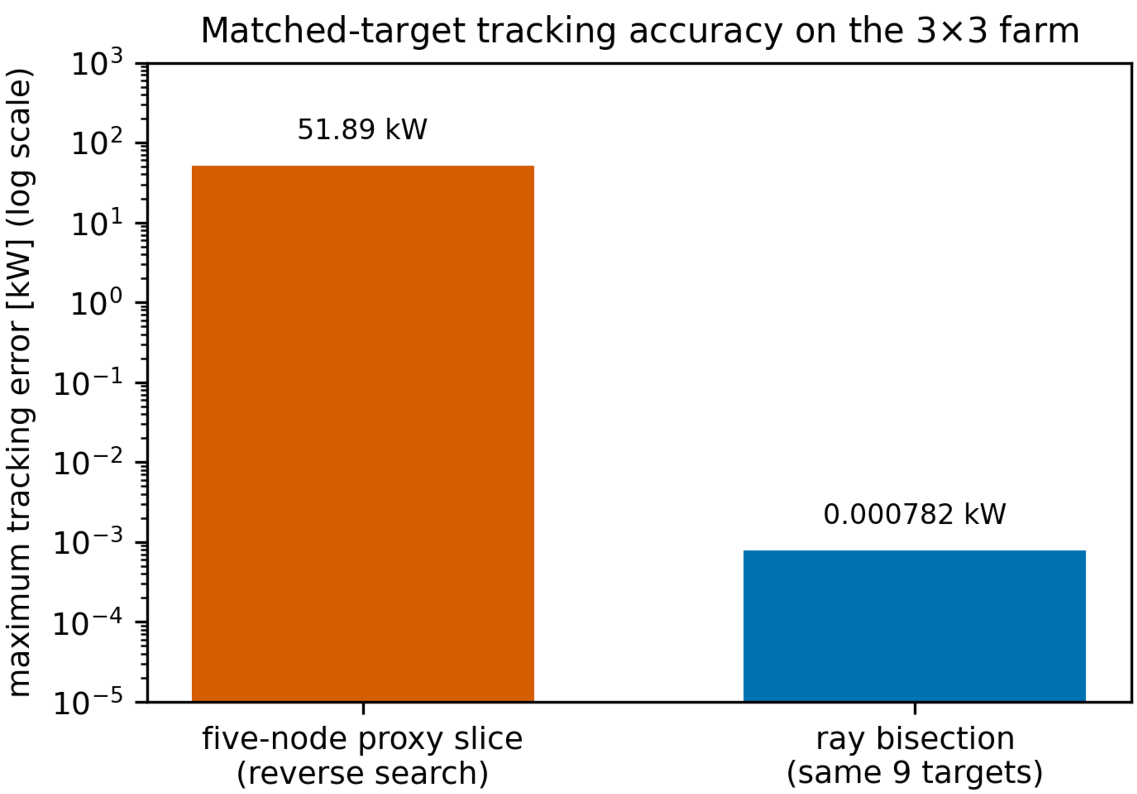

Figure 4: Matched-target proxy versus exact inverse on the 3×3 power range. Over the nine targets of Table 2, the five-node proxy slice with reverse search errs by up to 51.89 kW (0.52 % of $P_{\max}$); ray bisection reaches $\leq 7.8 \cdot 10^{-4}$ kW in 7–11 calls, a factor 6.6×10^4 lower maximum error.

6 Discussion

- **Field-scale remark.** At $8\,\mathrm{m\,s^{-1}}$ the 3×3 tracking range is $[8095, 10041]\,\mathrm{kW}$: an attainable band of 24 % of baseline, comparable to the curtailment ranges real farms bid into frequency services (Tamaro et al., 2025); the +7.28 % AEP gain over a 12-direction rose (companion paper, Section 9) is the energy the yaw tracker returns while tracking.

- **Extensions.** Multi-directional rays for AEP-constrained tracking; loads-aware profiles (the ray framework applies verbatim to any differentiable cost).

- **Experimental path.** Ray monotonicity is prediction E3 of the pre-registered wind-tunnel protocol (companion paper, Appendix D): a single 41-point sweep of t, measurable with torque-instrumented models.

- **Limitations.** Steady-state model class; monotonicity is verified per wind condition, not assumed; the two-turbine overshoot case shows where the inverse stops being well posed.

7 Conclusions

We have shown that the inverse problem of wake steering, historically avoided, becomes a solved special case once the second-order structure of the farm-power map is known. Power tracking reduces to bracketed bisection on a certified monotone ray; the inverse map is Lipschitz with computable constant, so target changes translate boundedly into profile changes; and the matched-target comparison with the bilinear-proxy tracker quantifies what the upgrade buys: a factor 6.6×10^4 (about 4.8 orders of magnitude) reduction in maximum error. The scheme needs 7–11 model calls per target on a 3×3 farm, comes with a well-posedness certificate, and composes with wind-direction AEP. Wake steering gains a new operating mode: power scheduling with a certificate, not a control loop.

A Reproduction

`exp_inverse.py` (ray monotonicity checks, Table 2, and the matched-target proxy comparison written to `expcache/table2_tracking.json` and `expcache/proxy_tracking_benchmark.json`), `make_figures2.py` (Figs. C1–C4, including the bisection budget study), `analytic_3chain.py` (closed-form ray derivatives for the three-turbine chain). FLORIS 4.6.6, `default_inputs.yaml`, $8\,\mathrm{m\,s^{-1}}$, TI 0.06, wd 270°, box $[0°, 30°]$. Rays sampled at 41 points; bisection tolerance 10^{-6} on the scale. Evaluations at $\gamma \geq 25°$ near the box corners produce negative-velocity warnings in FLORIS and are excluded from claims, matching the companion paper's validity boundary.

Acknowledgements

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